STA 2002

Lectures 11-12

Power

Sample Size

Reminder of the terminology

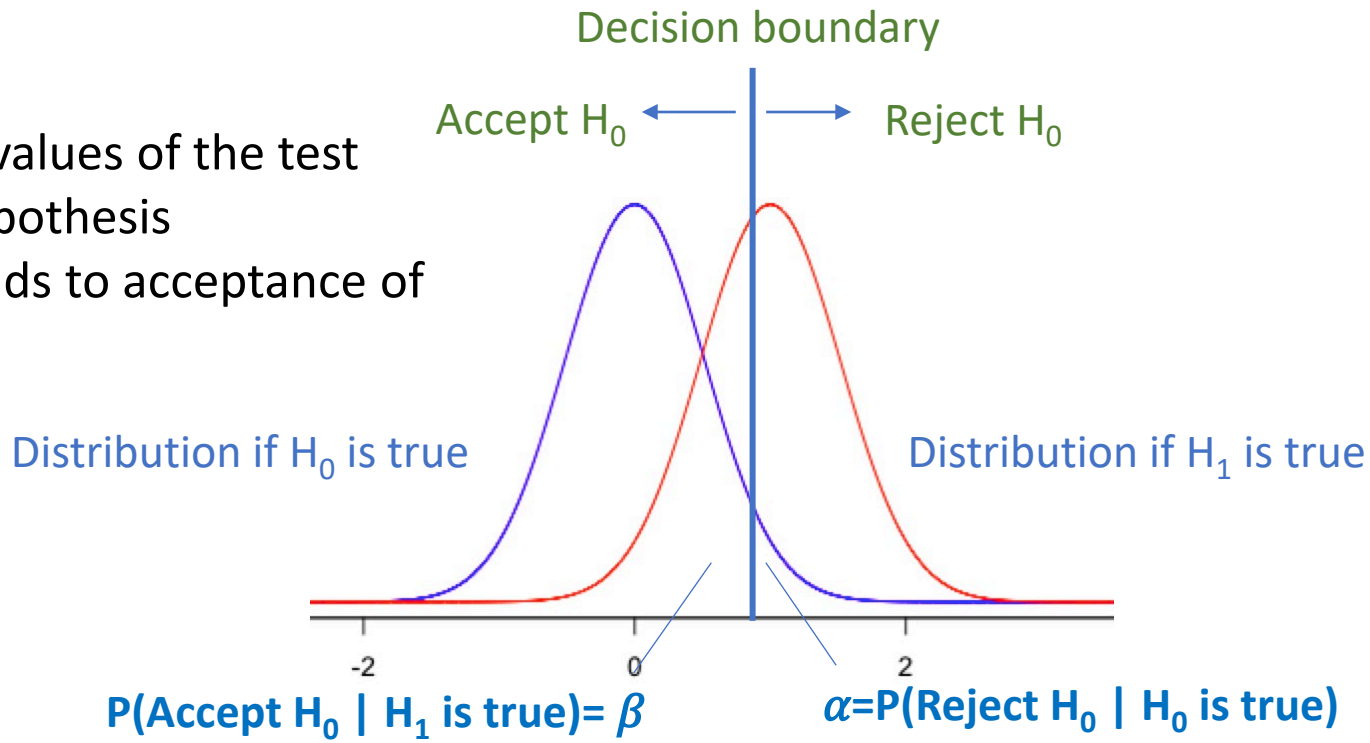
- **Type I error:** rejecting H_0 when it is true**.**
- **Significance level** (α): the probability of a type I error.

- **Type II error** (β): accepting the null hypothesis when it is false.
- **Power** of the test (1 − β): the probability that the null hypothesis is rejected when it is false.

These can only be calculated when the alternative hypothesis is a simple hypothesis.

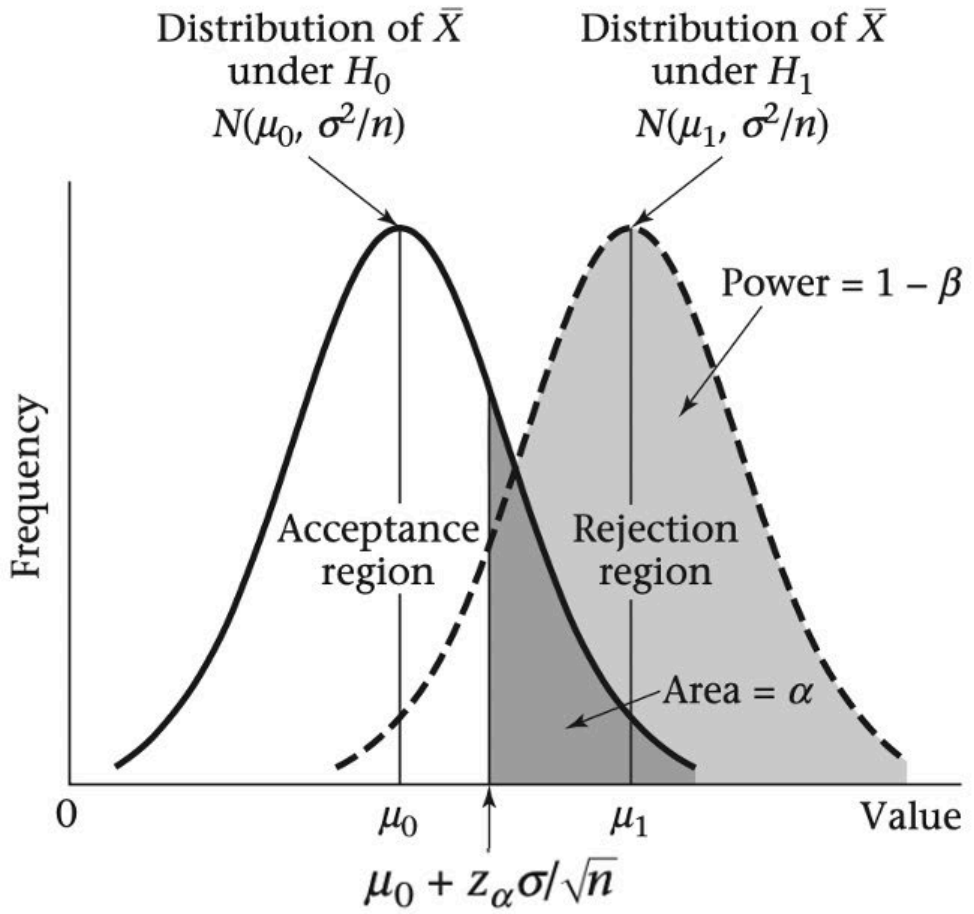
- **Null distribution:** the probability distribution of the test statistic when the null hypothesis is true

- **Rejection region (critical region):** the set of values of the test statistic that leads to rejection of the null hypothesis
- **Acceptance region:** the set of values that leads to acceptance of the null hypothesis



One-sample, one-sided test of the mean with known σ

Illustration of power for the one-sample test for the mean of a normal distribution with known variance ($\mu_1 > \mu_0$)



$$\text{Power} = Pr\left(\text{reject } H_0 \middle| H_0 \text{ false}\right)$$

$$= Pr\left(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} > z_\alpha \mid \mu = \mu_1\right)$$

$$= Pr\left(\bar{X} > \mu_0 + z_\alpha\, \sigma/\sqrt{n} \mid \mu = \mu_1\right)$$

$$= Pr\left(\frac{\bar{X}-\mu_1}{\sigma/\sqrt{n}} > \frac{\mu_0-\mu_1}{\sigma/\sqrt{n}} + z_\alpha \mid \mu = \mu_1\right)$$

$$= Pr\left(Z^* > \frac{\mu_0-\mu_1}{\sigma/\sqrt{n}} + z_\alpha \mid \mu = \mu_1\right)$$

$$= Pr\left(Z^* < \frac{\mu_1-\mu_0}{\sigma/\sqrt{n}} - z_\alpha \mid \mu = \mu_1\right)$$

$$= Pr\left(Z^* < \frac{\mu_1-\mu_0}{\sigma/\sqrt{n}} + z_{1-\alpha} \mid \mu = \mu_1\right)$$

The power is: $\Phi\left[z_{1-\alpha} + \frac{\mu_1 - \mu_0}{\sigma}\sqrt{n}\right] = \Phi\left[-z_\alpha + \frac{\mu_1 - \mu_0}{\sigma}\sqrt{n}\right]$

One-sample, one-sided test of the mean with known σ

Example : Metro EMS

A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating in a multiple hospital system with approximately 20 mobile medical units, the service goal is to respond to medical emergencies with a mean time of 12 minutes or less. The director of medical services wants to formulate a hypothesis test that could use a sample of emergency response times to determine whether or not the service goal of 12 minutes or less is being achieved. To test these hypotheses, a random sample of 40 past emergency response records were examined. The resulting mean response time was 13.25 minutes, assuming the true standard deviation equal to 3.2 minutes. The level of significance is 0.05. We want to get the power for $\mu_1 = 13.5\ min$.

Solution:

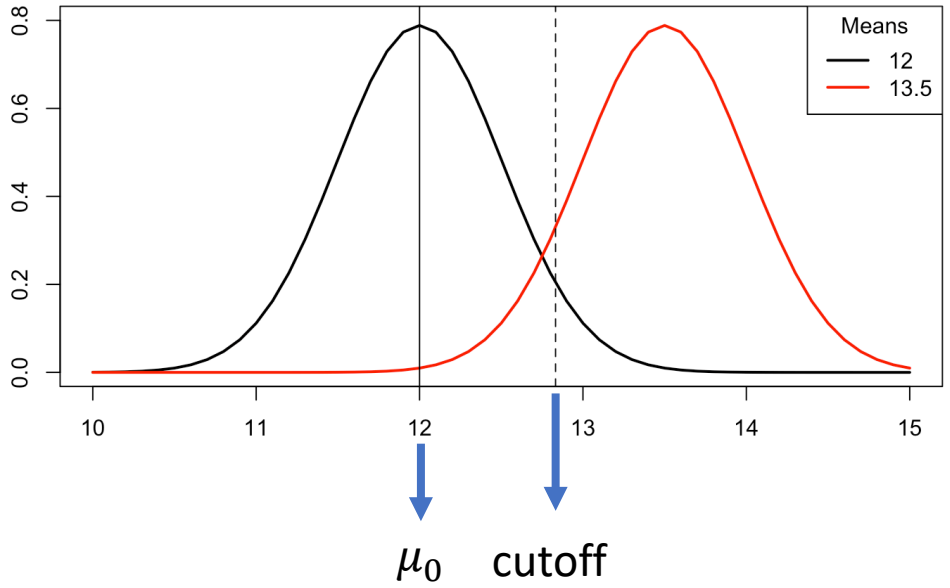
Let μ denote the mean emergency response time. The hypotheses are $H_0: \mu \leq 12$ and $H_a: \mu > 12$.

Let $\mu_0 = 12\ min, \mu_1 = 13.5\ min$, we have $\alpha = .05,\ \sigma = 3.2,\ n = 40$

$$\text{Power} = \phi\left[-z_\alpha + \frac{\mu_1-\mu_0}{\sigma}\sqrt{n}\right] = \phi\left[-1.645 + \frac{13.5-12}{3.2}\sqrt{40}\right]$$

$$= \phi(1.3196) = .907$$

```
> pnorm(1.3196)
[1] 0.9065157
```



Therefore, if the alternative hypothesis is true, there is about 90.7% chance of detecting a significant difference using a 5% significance level with this sample size.

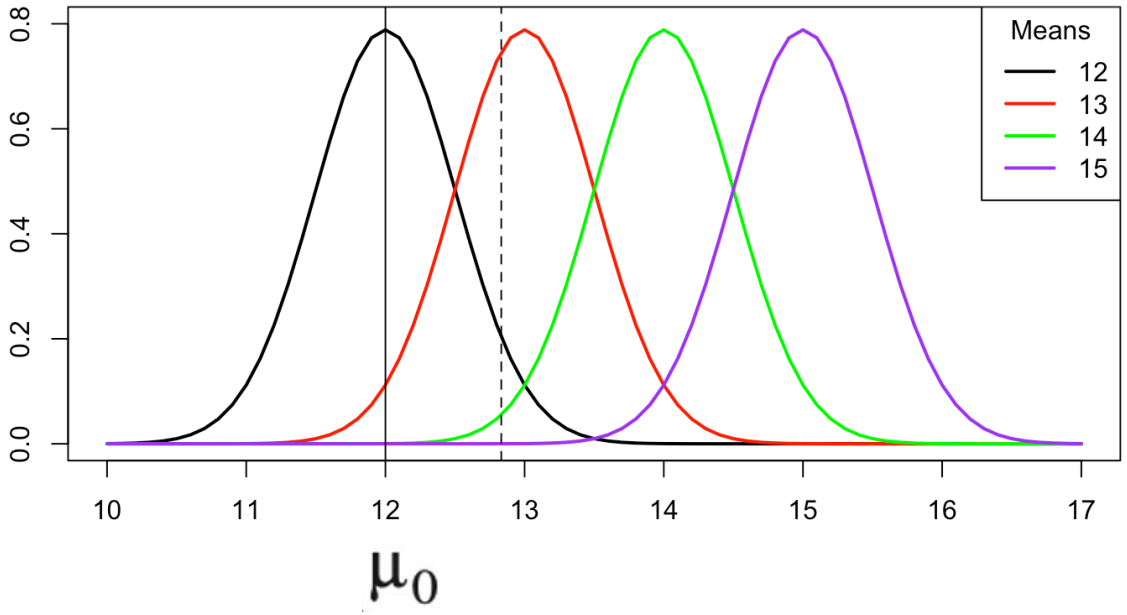
What if $\mu_1 = 12, 13, 14, 15\ min$?

One-sample, one-sided test of the mean with known σ

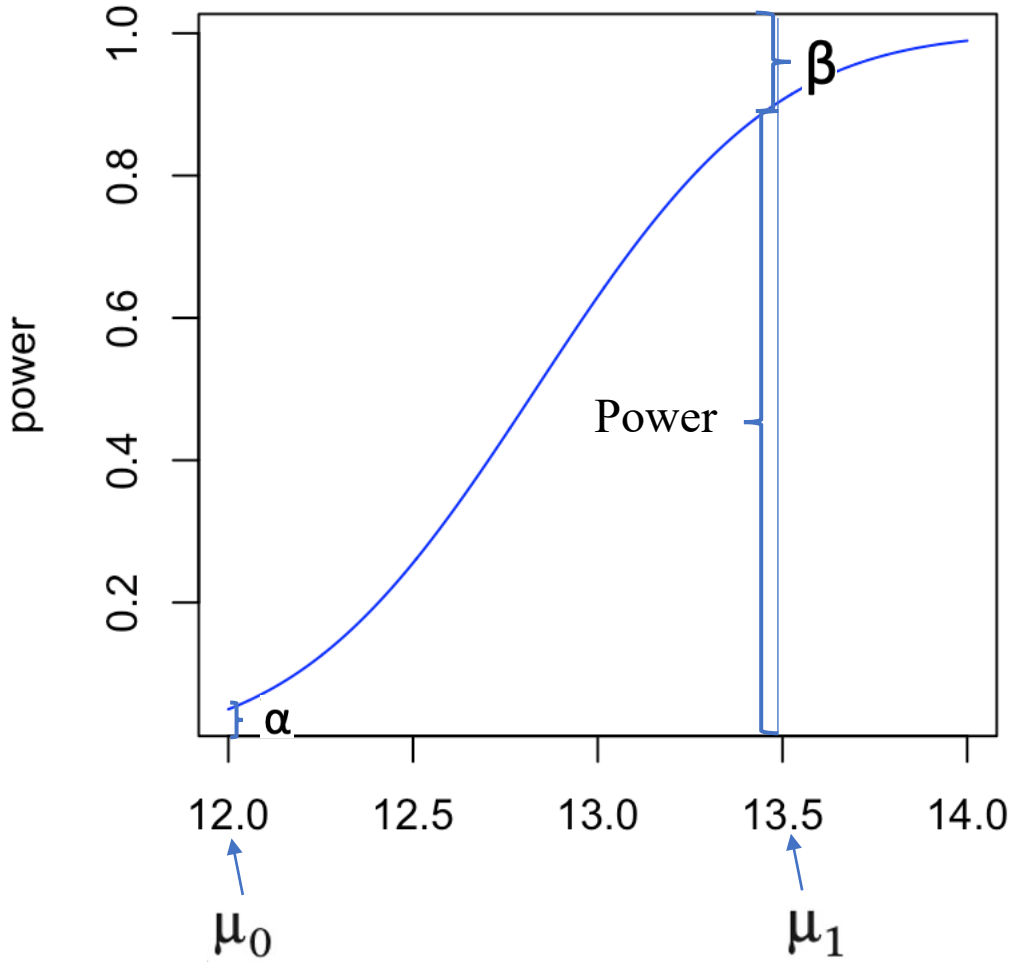
$$\Phi\left[z_{1-\alpha} + \frac{\mu_1 - \mu_0}{\sigma}\sqrt{n}\right]$$

Power function of the test

The probability $K(\mu)$ of rejecting H_0

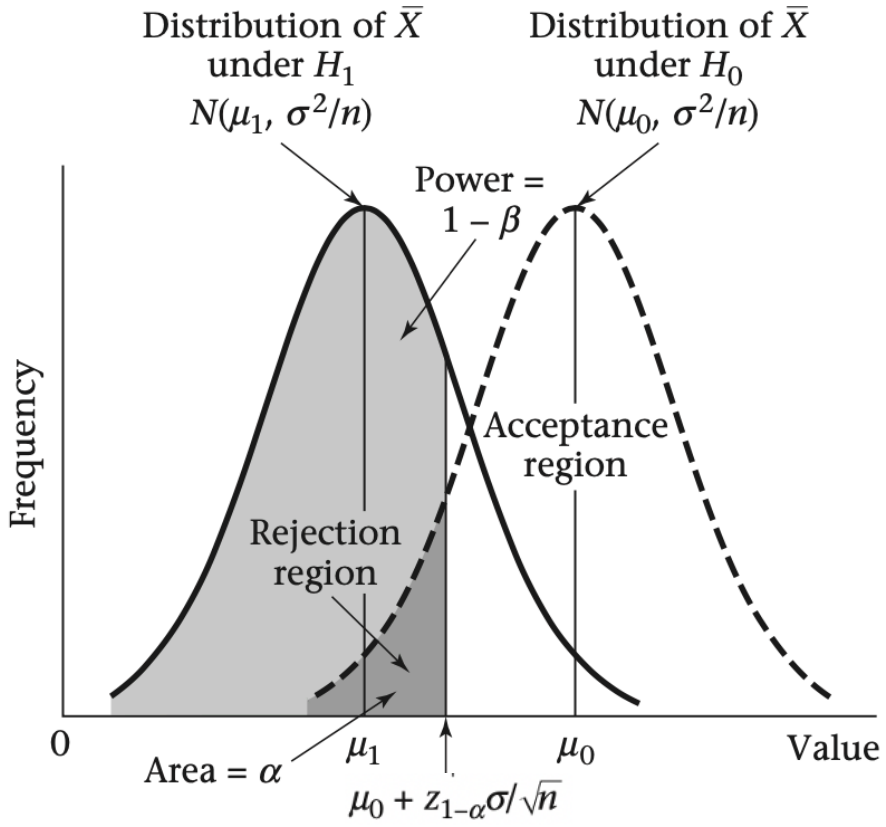


Power function for the one-sided significance test for the response time data in the Metro EMS example



One-sample, one-sided test of the mean with known σ

Illustration of power for the one-sample test for the mean of a normal distribution with known variance ($\mu_1 < \mu_0$)



$$\text{Power} = Pr\left(\text{reject } H_0 \,|\, H_0 \text{ false}\right)$$

$$= Pr\left(\frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}} < z_{1-\alpha} \,|\, \mu = \mu_1\right)$$

$$= Pr\left(\bar{X} < \mu_0 + z_{1-\alpha}\,\sigma/\sqrt{n} \,|\, \mu = \mu_1\right)$$

$$= Pr\left(\frac{\bar{X}-\mu_1}{\sigma/\sqrt{n}} < \frac{\mu_0-\mu_1}{\sigma/\sqrt{n}} + z_{1-\alpha} \,|\, \mu = \mu_1\right)$$

$$= Pr\left(Z^* < \frac{\mu_0-\mu_1}{\sigma/\sqrt{n}} + z_{1-\alpha} \,|\, \mu = \mu_1\right)$$

For one-sided test, power is: $\Phi\left[z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma}\sqrt{n}\right] = \Phi\left[-z_\alpha + \frac{\mu_0 - \mu_1}{\sigma}\sqrt{n}\right]$

One-sample, one-sided test of the mean with known σ

Example : Metro EMS

Suppose that we are instead testing the following hypotheses:

$H_0: \mu \geq 12$ and $H_a: \mu < 12$

We want to get the power for $\mu_1 = 11\ min$.

Solution:

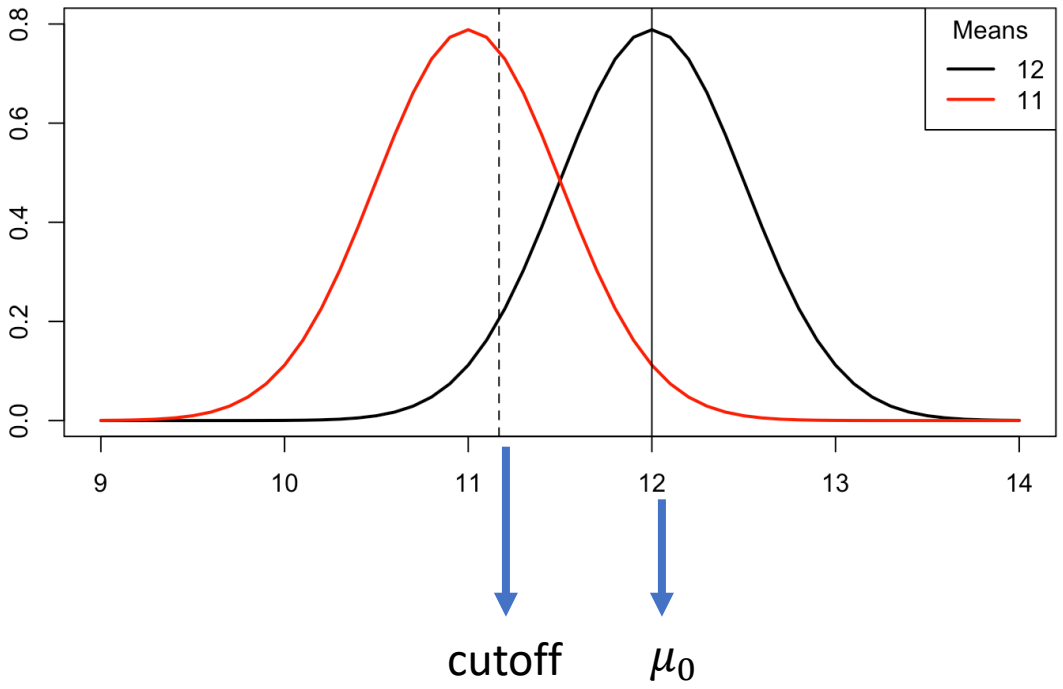
We have $\mu_0 = 12\ min, \mu_1 = 11\ min, \alpha = .05,\ \sigma = 3.2,\ n = 40$

$$\text{Power} = \Phi\left[z_{1-\alpha} + \frac{\mu_0 - \mu_1}{\sigma}\sqrt{n}\right] = \Phi\left[-1.645 + \frac{12-11}{3.2}\sqrt{40}\right]$$

$$= \Phi(0.33) = 0.6293$$

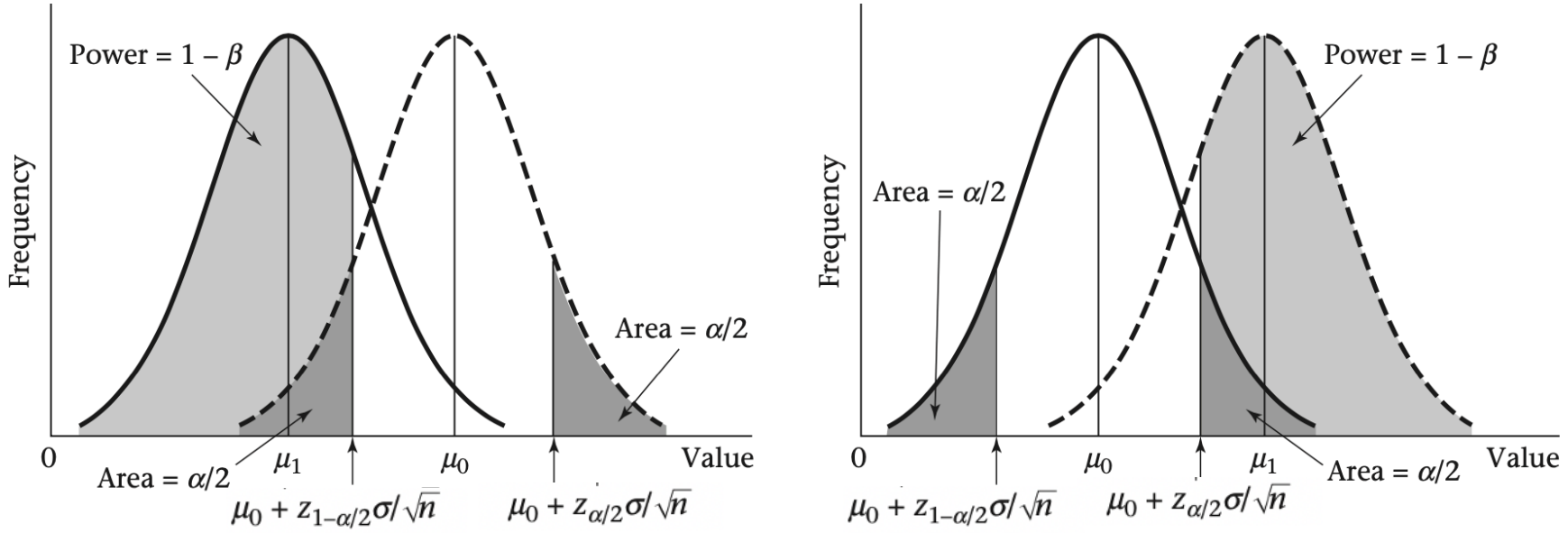
The chance of finding a significance difference is 62.93%

```
> pnorm(0.33)
[1] 0.6293
```



One-sample, two-sided test of the mean with known σ

Illustration of power for a two-sided test for the mean of a normal distribution with known variance



(a) $\mu_1 < \mu_0$ (b) $\mu_1 > \mu_0$

Power for the One-Sample *z* Test for the Mean of a Normal Distribution (Two-Sided Alternative)

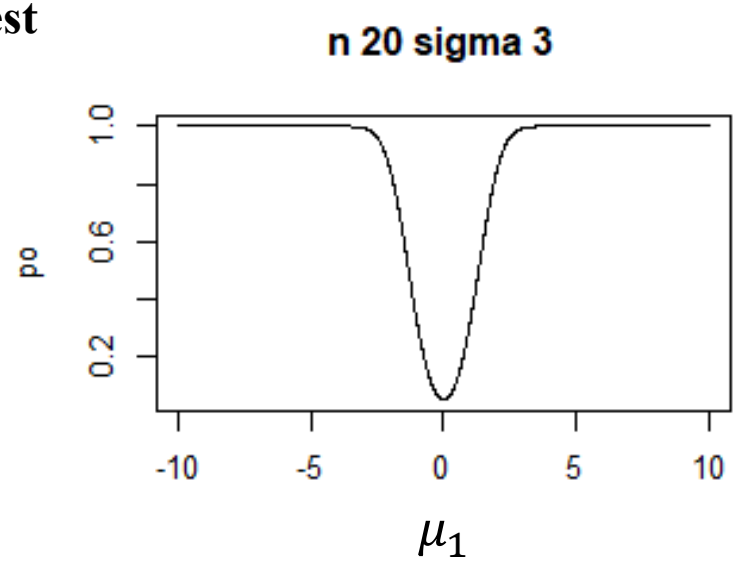
The power of the two-sided test H_0: $\mu = \mu_0$ vs. H_1: $\mu \neq \mu_0$ for the specific alternative $\mu = \mu_1$, where the underlying distribution is normal and the population variance (σ^2) is assumed known, is given exactly by

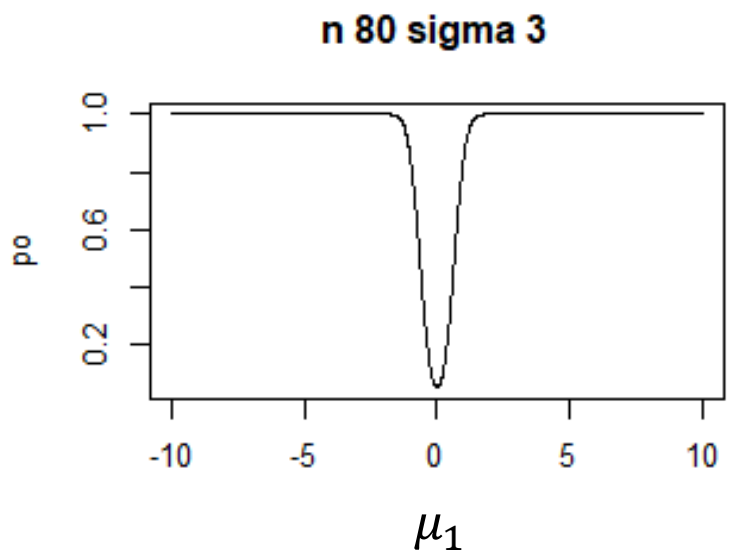
$$\Phi\left[-z_{\alpha/2} + \frac{(\mu_0 - \mu_1)\sqrt{n}}{\sigma}\right] + \Phi\left[-z_{\alpha/2} + \frac{(\mu_1 - \mu_0)\sqrt{n}}{\sigma}\right]$$

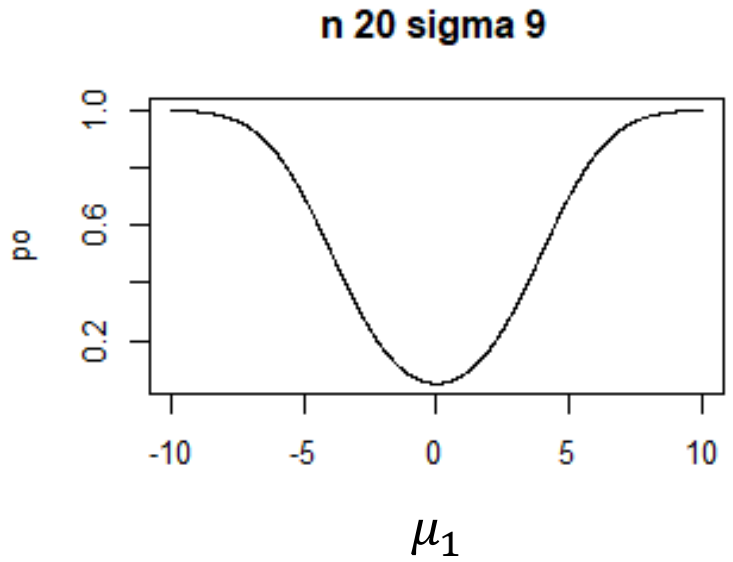
One-sample, two-sided test of the mean with known σ

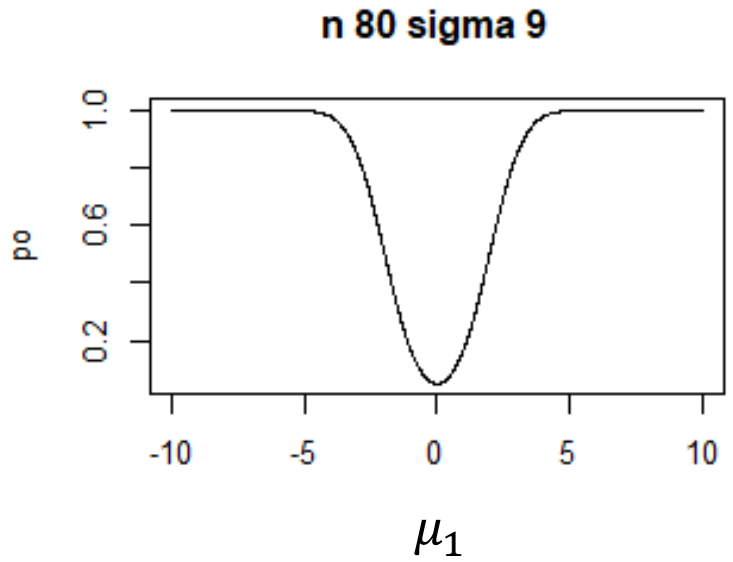
$$\Phi\left[-z_{\alpha/2}+\frac{(\mu_0-\mu_1)\sqrt{n}}{\sigma}\right]+\Phi\left[-z_{\alpha/2}+\frac{(\mu_1-\mu_0)\sqrt{n}}{\sigma}\right]$$

Power function of the test









One-sample, two-sided test of the mean with known σ

Example: Glow Toothpaste

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces. Periodically, a sample of 30 tubes will be selected in order to check the filling process. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted. Assume that a sample of 30 toothpaste tubes provides a sample mean of 6.1 ounces. We assume the population standard deviation is 0.2 ounces.

What power would such quality assurance procedures have of detecting a significant difference in the mean of the tubes if it is hypothesized that the true mean of the tube could be either an increase or a decrease of of 0.5 ounces?

Solution

$\sigma = 0.2, |\mu_0 - \mu_1| = 0.5, \alpha = .05, n = 30.$ We have

$$Power = \Phi(-z_{\frac{0.05}{2}} + 0.5 * \frac{\sqrt{30}}{0.2}) + \Phi\left(-z_{\frac{0.05}{2}} - 0.5 * \frac{\sqrt{30}}{0.2}\right) \approx \Phi(-1.96 + 13.69) \approx 1$$

The quality assurance procedures would be almost certain of detecting a significant difference

Factors affecting power

Factors Affecting the Power

(1) If the significance level is made smaller (α decreases), z_α increases and hence the power decreases.

(2) If the alternative mean is shifted farther away from the null mean ($|\mu_0 - \mu_1|$ increases), then the power increases.

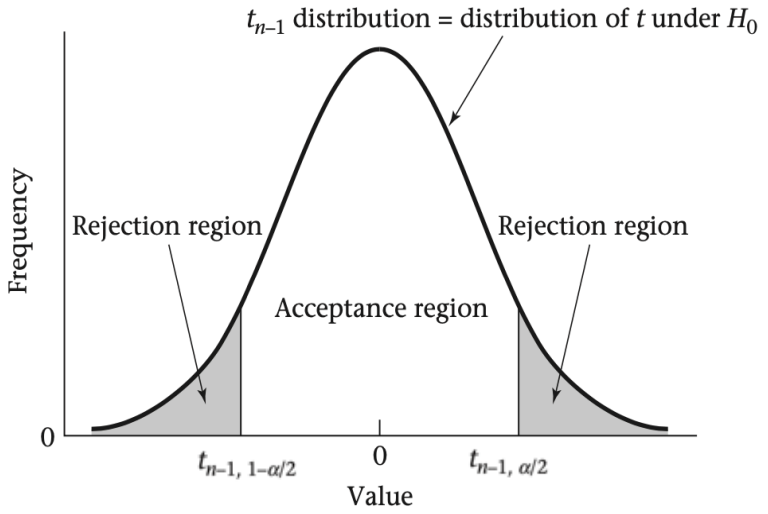
(3) If the standard deviation of the distribution of individual observations increases (σ increases), then the power decreases.

(4) If the sample size increases (n increases), then the power increases.

Why do we assume we know σ?

Not required

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$$



The threshold values involve S which is a random variable, e.g., you need to calculate $Pr\ (T^* < \frac{\mu_0 - \mu_1}{S/\sqrt{n}} + t_{1-\alpha} | \ \mu = \mu_1)$!

The same μ_1 means different power with different S.

We have to assume something about the variance.

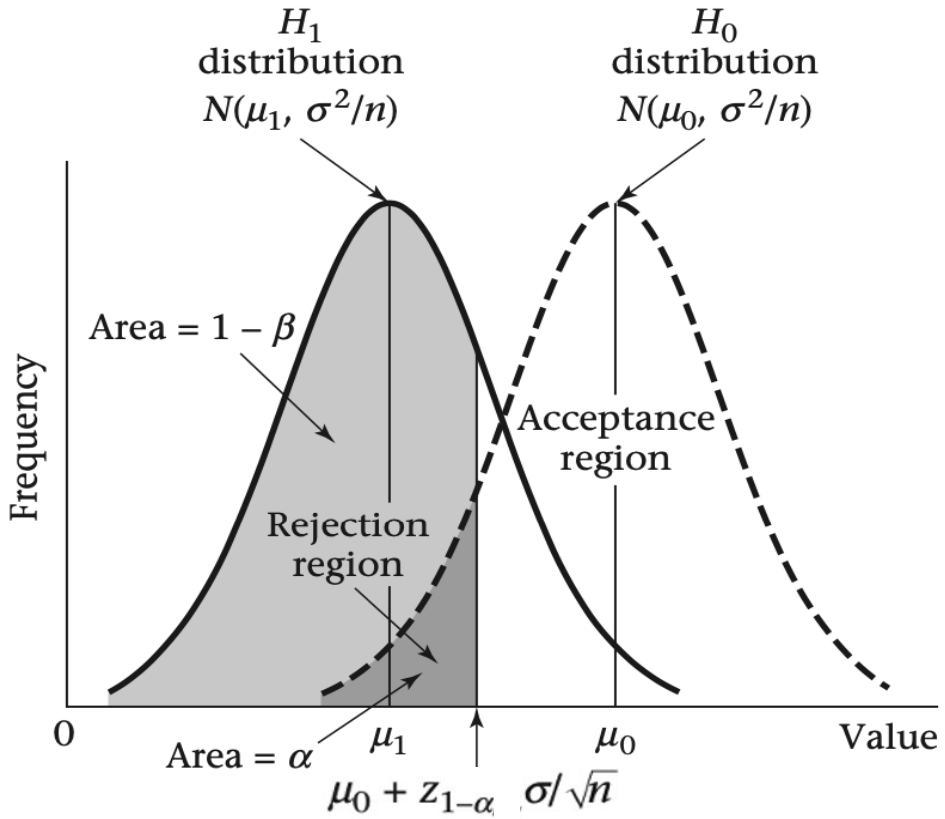
A common way to work around if we don't know σ is to use a relative effect size: $\frac{\Delta}{\sigma}$

Sample size for one-sided test of the mean with known σ

A one-sided significance test will be conducted at level α
The true alternative mean is **assumed** to be μ_1,

What sample size is needed to be able to detect a significant difference (reject H_0: $\mu = \mu_0$) with probability $1-\beta$?

Requirements for appropriate sample size



$$\text{Power} = \Phi\left(z_{1-\alpha} + |\mu_0 - \mu_1|\sqrt{n}/\sigma\right) = 1 - \beta$$

$$z_{1-\alpha} + |\mu_0 - \mu_1|\sqrt{n}/\sigma = z_\beta$$

$$\sqrt{n} = \frac{\left(-z_{1-\alpha} + z_\beta\right)\sigma}{|\mu_0 - \mu_1|}$$

$$n = \frac{\left(z_\alpha + z_\beta\right)^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

Sample size for one-sample, one-sided test of the mean with known σ

Example : Metro EMS

Consider the Metro EMS example: A major west coast city provides one of the most comprehensive emergency medical services in the world. Operating in a multiple hospital system with approximately 20 mobile medical units, the service goal is to respond to medical emergencies with a mean time of 12 minutes or less. The director of medical services wants to formulate a hypothesis test that could use a sample of emergency response times to determine whether or not the service goal of 12 minutes or less is being achieved. Suppose that $\mu_0 = 12$, $\mu_1 = 12.5$, $\sigma = 3.2$, $\alpha = .05$, $1 - \beta = .80$, and we use a one-sided test. Compute the appropriate sample size needed to conduct the test.

Solution:

$$n = \frac{(z_\alpha + z_\beta)\sigma^2}{(\mu_0 - \mu_1)^2} = \frac{(z_{0.05} + z_{0.2})3.2^2}{(12 - 12.5)^2}$$

$$= \frac{(z_{0.05} + z_{0.2})3.2^2}{(12 - 12.5)^2} = \frac{(1.645 + 0.842)3.2^2}{(12 - 12.5)^2} = 101.9$$

Always round up to ensure sufficient power, so round to 102.

Derive $z_{0.05}$, $z_{0.2}$ in R:

```
> qnorm(0.95)
[1] 1.644854
> qnorm(0.8)
[1] 0.8416212
```

$$n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_0 - \mu_1)^2}$$

Factors affecting sample size

Factors Affecting the Sample Size

(1) The sample size increases as σ^2 increases.

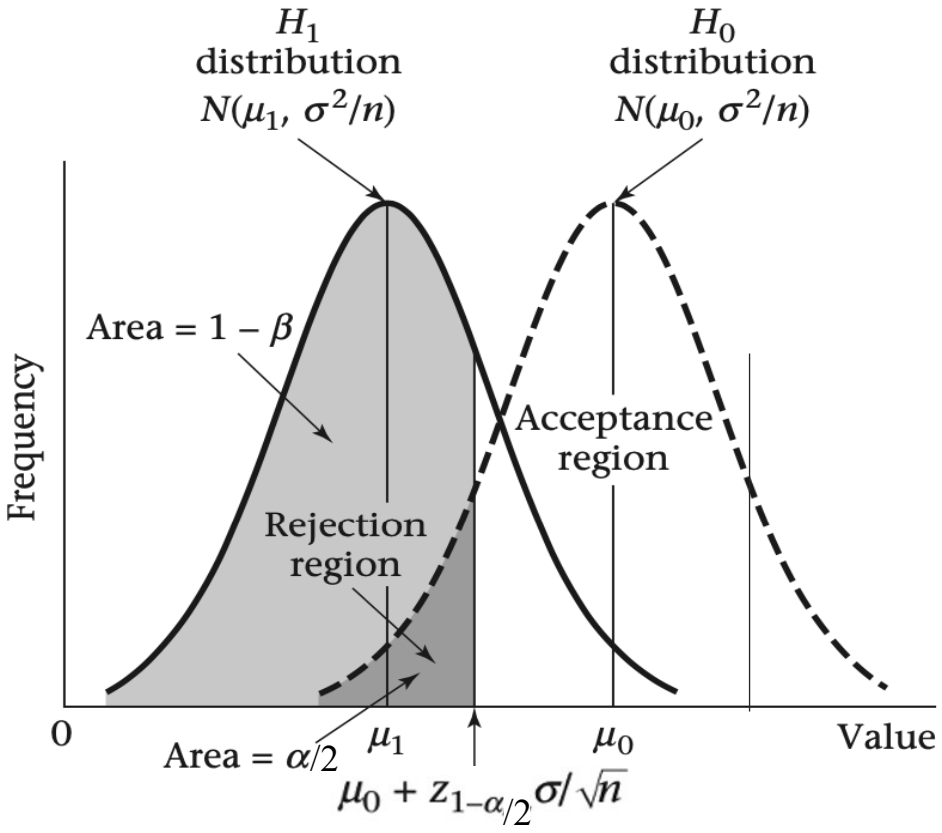
(2) The sample size increases as the significance level is made smaller (α decreases).

(3) The sample size increases as the required power increases ($1 - \beta$ increases).

(4) The sample size decreases as the absolute value of the distance between the null and alternative means ($|\mu_0 - \mu_1|$) increases.

Sample size for one-sample, two-sided test of the mean with known σ

Requirements for appropriate sample size



$$n = \frac{\sigma^2\left(z_\beta + z_{\alpha/2}\right)^2}{\left(\mu_0 - \mu_1\right)^2}$$

Sample size for one-sample, two-sided test of the mean with known σ

For two-sided alternative,

$$n = \frac{\sigma^2 \left(z_\beta + z_{\alpha/2} \right)^2}{(\mu_0 - \mu_1)^2}$$

Example: Glow Toothpaste

The production line for Glow toothpaste is designed to fill tubes of toothpaste with a mean weight of 6 ounces. Quality assurance procedures call for the continuation of the filling process if the sample results are consistent with the assumption that the mean filling weight for the population of toothpaste tubes is 6 ounces; otherwise the filling process will be stopped and adjusted.
Suppose we want at least 80% power for detecting a significant difference if the filling weight is 0.1 ounces higher or lower and the standard deviation is 0.2 ounces. How many tubes should we sample for the test?

We assume $\sigma = 0.2, \alpha = .05, 1 - \beta = .80,$ and we use a two-sided test because we are not sure in what direction that filling weight will change

$$n = \frac{(z_{\alpha/2} + z_\beta)\sigma^2}{(|\mu_0 - \mu_1|)^2} = \frac{(z_{0.025} + z_{0.2})0.2^2}{(0.1)^2}$$

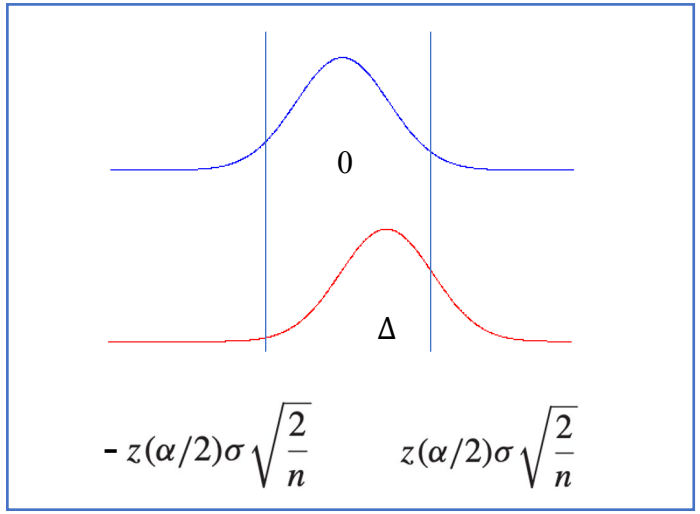
$$= \frac{(1.96 + 0.842)0.2^2}{(0.1)^2} = 11.2, \text{ or } 12 \text{ samples}$$

Obtain $z_{0.025}, and\ z_{0.2}$ in R:

```
> qnorm(0.975)
[1] 1.959964
> qnorm(0.8)
[1] 0.8416212
```

Power and sample size for two-sample, two-sided test of the mean with known σ

$H_0: \mu_X = \mu_Y$ $\quad$ $H_1: \mu_X \neq \mu_Y$



Suppose equal sample size n,

$$\mathrm{Var}(\overline{X} - \overline{Y}) = \sigma^2\left(\frac{1}{n} + \frac{1}{n}\right) = \frac{2\sigma^2}{n}$$

test statistic $\quad Z = \dfrac{\overline{X} - \overline{Y}}{\sigma\sqrt{2/n}}$

The rejection region

$$|Z| > z_{\alpha/2}$$

$$|\overline{X} - \overline{Y}| > z_{\alpha/2}\,\sigma\sqrt{\frac{2}{n}}$$

if $\mu_X - \mu_Y = \Delta$

$$\text{Power} = P\left[|\overline{X} - \overline{Y}| > z_{\alpha/2}\,\sigma\sqrt{\frac{2}{n}}\right]$$

$$= P\left[\overline{X} - \overline{Y} > z_{\alpha/2}\,\sigma\sqrt{\frac{2}{n}}\right] + P\left[\overline{X} - \overline{Y} < -z_{\alpha/2}\,\sigma\sqrt{\frac{2}{n}}\right]$$

$$P\left[\overline{X} - \overline{Y} > z_{\alpha/2}\sigma\sqrt{\frac{2}{n}}\right] = P\left[\frac{(\overline{X} - \overline{Y}) - \Delta}{\sigma\sqrt{2/n}} > \frac{z_{\alpha/2}\sigma\sqrt{2/n} - \Delta}{\sigma\sqrt{2/n}}\right] = 1 - \Phi\left[z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right]$$

$$P\left[\overline{X} - \overline{Y} < -z_{\alpha/2}\sigma\sqrt{\frac{2}{n}}\right] = \Phi\left[-z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right]$$

$$\text{Power} = 1 - \Phi\left[z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right] + \Phi\left[-z_{\alpha/2} - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right]$$

Power and sample size for two-sample, two-sided test of the mean with known σ

Assume that we have samples from two normal distributions whose standard deviations are both 5
The null hypothesis is tested at a significance level of .05.

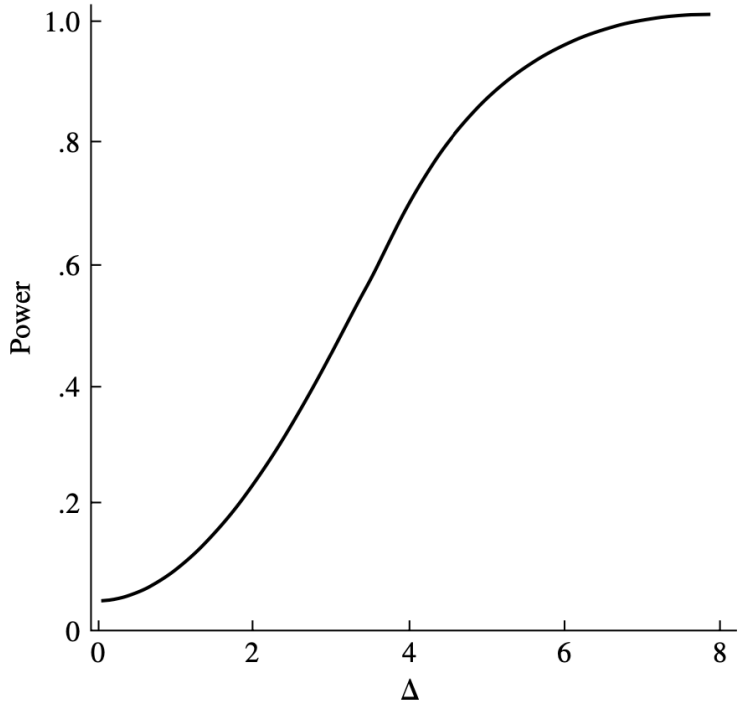
Suppose that we wanted to be able to detect a difference of $\Delta = 1$ with probability 0.9. What sample size would be necessary?

Using only the dominant term (the other is pretty much zero)

$$\Phi\left(1.96 - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}}\right) = .1$$

$$1.96 - \frac{\Delta}{\sigma}\sqrt{\frac{n}{2}} = -1.28$$

$$n = 525$$



Plot of power versus Δ for samples of size 18

Power and sample size for one-sample test of proportion

Finding power using normal approximation.

$H_0: p = 0.5$
$H_1: p \neq 0.5$

What power can we achieve if $p=0.25$, $\alpha=0.05$, and n=30?

Under the null: $x \sim N(np_0, np_0(1-p_0)) = \mathrm{N}(15, 7.5)$

Rejection region: $x < 15 - 1.96 \times 2.74 \; or \; x > 15 + 1.96 \times 2.74$

$x < 9.63 \; or \; x > 20.37$

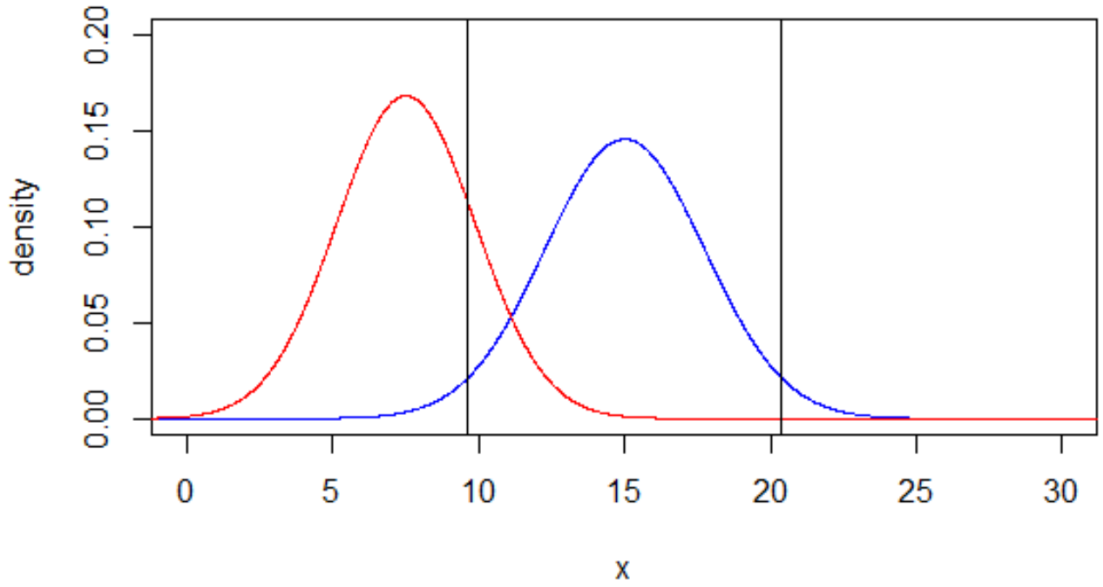
When $p=0.25$, $x \sim N(np, np(1-p)) = \mathrm{N}(7.5, 5.625)$

$x < 9.63 \; or \; x > 20.37$

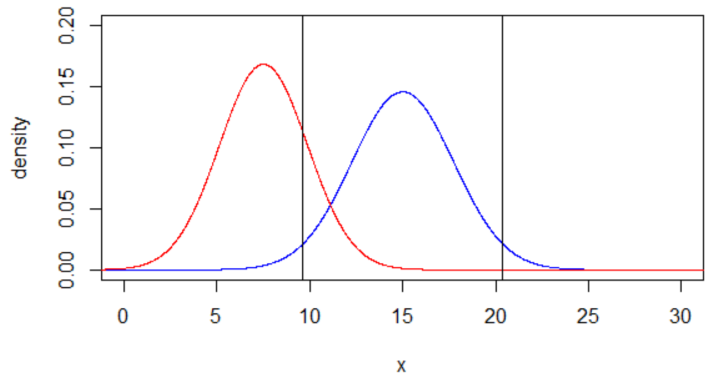
$z < \frac{9.63-7.5}{2.37} \; or \; z > \frac{20.37-7.5}{2.37}$

$z < 0.899 \; or \; z > 5.43$

$\Phi(0.899) = 0.815$



Power and sample size for one-sample test of proportion



Finding sample size using normal approximation.

$H_0: p = 0.5$
$H_1: p \neq 0.5$

What sample size will give us 80% power to detect *p=0.25* at alpha level of 0.05?

$x \sim N(np_0, np_0(1-p_0))$
$x \sim N(n/2, n/4)$

$$Rejection\ region: \frac{x - n/2}{\sqrt{n/4}} < -1.96 \cup \frac{x - n/2}{\sqrt{n/4}} > 1.96$$

$$x < n/2 - 0.98\sqrt{n} \cup x > n/2 + 0.98\sqrt{n}$$

Under *p=0.25* ,

$x \sim N(n/4, 3n/16)$

$$z < \frac{n/2 - 0.98\sqrt{n} - n/4}{\sqrt{3n/16}} = \frac{n/4 - 0.98\sqrt{n}}{\sqrt{3n/16}} = \frac{\sqrt{n}}{\sqrt{3}} - 2.263$$

$$\frac{\sqrt{n}}{\sqrt{3}} - 2.263 = \Phi^{-1}(0.8) = 0.84$$

$$n = 28.9 \longrightarrow \text{n=29}$$

$$P\left(z > \frac{\sqrt{n}}{\sqrt{3}} + 2.263\right) \cong 0$$

When n is reasonably big. Ignored.

Power and sample size for one-sample test of proportion

Not required

$$x \sim N(np_0, np_0(1-p_0))$$

Sample size n,

$H_0: p = p_0$
$H_1: p \neq p_0$

$$Rejection\ region: \frac{x - np_0}{\sqrt{np_0(1-p_0)}} < -z_{\alpha/2} \cup \frac{x - np_0}{\sqrt{np_0(1-p_0)}} > z_{\alpha/2}$$

$$x < np_0 - z_{\alpha/2}\sqrt{np_0(1-p_0)} \cup x > np_0 + z_{\alpha/2}\sqrt{np_0(1-p_0)}$$

If true proportion is q,

$$Prob\left(x < np_0 - z_{\alpha/2}\sqrt{np_0(1-p_0)} \middle| q\right) = \Phi\left(\frac{np_0 - z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right)$$

$$Prob\left(x > np_0 + z_{\alpha/2}\sqrt{np_0(1-p_0)} \middle| q\right) = 1 - \Phi\left(\frac{np_0 + z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right)$$

$$Prob(reject\ H_0|q) = 1 - \Phi\left(\frac{np_0 + z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right) + \Phi\left(\frac{np_0 - z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right)$$

$$= 1 - \Phi\left(\frac{n(p_0 - q) + z_{\alpha/2}\sqrt{np_0(1-p_0)}}{\sqrt{nq(1-q)}}\right) + \Phi\left(\frac{n(p_0 - q) - z_{\alpha/2}\sqrt{np_0(1-p_0)}}{\sqrt{nq(1-q)}}\right)$$

Power and sample size for one-sample test of proportion

Not required

$$Prob(reject\ H_0|q)$$

$$= 1 - \Phi\left(\frac{np_0 + z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right) + \Phi\left(\frac{np_0 - z_{\alpha/2}\sqrt{np_0(1-p_0)} - nq}{\sqrt{nq(1-q)}}\right)$$

$$= 1 - \Phi\left(\frac{n(p_0-q) + z_{\alpha/2}\sqrt{np_0(1-p_0)}}{\sqrt{nq(1-q)}}\right) + \Phi\left(\frac{n(p_0-q) - z_{\alpha/2}\sqrt{np_0(1-p_0)}}{\sqrt{nq(1-q)}}\right)$$

$$= 1 - \Phi\left(\frac{\sqrt{n}(p_0-q)}{\sqrt{q(1-q)}} + z_{\alpha/2}\sqrt{\frac{p_0(1-p_0)}{q(1-q)}}\right) + \Phi\left(\frac{\sqrt{n}(p_0-q)}{\sqrt{q(1-q)}} - z_{\alpha/2}\sqrt{\frac{p_0(1-p_0)}{q(1-q)}}\right)$$

Usually dominated by one side of the alternative distribution

$$\cong \Phi\left(\frac{\sqrt{n}|p_0-q|}{\sqrt{q(1-q)}} - z_{\alpha/2}\sqrt{\frac{p_0(1-p_0)}{q(1-q)}}\right) = \Phi\left(\sqrt{\frac{p_0(1-p_0)}{q(1-q)}}\left(\frac{\sqrt{n}|p_0-q|}{\sqrt{p_0(1-p_0)}} - z_{\alpha/2}\right)\right)$$

How about a two-sample comparison, and consider a $\Delta = p_1 - p_2$?

Not required

Distribution of $\hat{p}_1 - \hat{p}_2 = \frac{x_1}{n_1} - \frac{x_2}{n_2}$	$\hat{p}_1 - \hat{p}_2 \sim N(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2})$

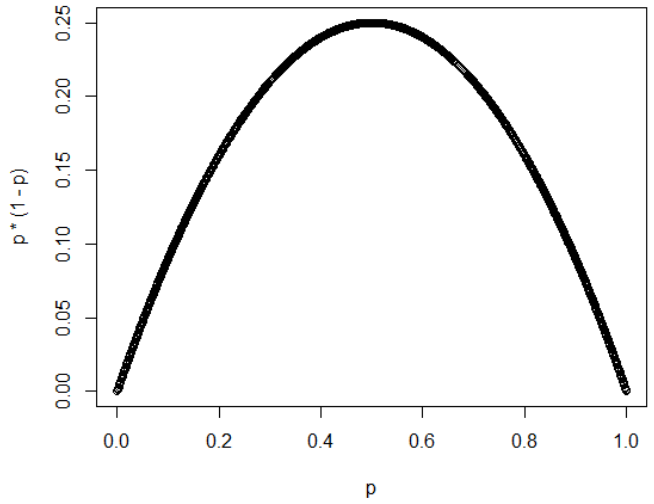
Even if we pre-specify Δ , the variance term still depends on p_1

So we cannot find power or sample size without specifying p_1

Certainly we can always do

Specify p_1,Δ, n → Find power

Specify p_1,Δ, β → Find n



In real studies, this almost always lead to plots of power/sample size against p_1 and Δ, and often use exact computation.

We ignore the details here.

The bottom line

(1) Think about the null distribution:

Find the critical value(s) assuming null is true.

(2) Think about the assumed simple alternative:

Find where the critical values (found in step 1) fall onto the alternative test statistic distribution.

- In calculation of power, the above is straight-forward, involving only numerical calculations.
- In calculation of sample size, the critical values involve “n”, which need to be solved in the second step.